Q5 Summary

A Comparative Evaluation of RAG Architectures for Cross-Domain LLM Applications: Design, Implementation, and Assessment. P. Ersoy and M. Erşahin (Commencis Teknoloji, Istanbul), IEEE Access, vol. 13, 2025, doi:10.1109/ACCESS.2025.3632404

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1. Problem and motivation

Large language models generalise well, but their knowledge is frozen at training time, and they sometimes invent specifics. In finance and e-commerce that is unacceptable: answers depend on exact figures, currencies and dates, policies change often, and regulators expect every claim to be traceable to a source. Retrieval-Augmented Generation (RAG) addresses this by retrieving evidence at query time and conditioning the answer on it. Four retrieval paradigms dominate practice (sparse, dense, hybrid and fusion), but earlier work evaluates them on different data, models and metrics. Practitioners therefore lack a like-for-like comparison of their accuracy, speed and auditability. The paper sets out to provide that comparison with a reproducible protocol.

The authors ask five research questions: (1) how the four pipelines compare on faithfulness and recall when the generator is held fixed; (2) how sensitive results are to retrieval depth, chunking and index tuning; (3) which paradigm gives the best utility under latency constraints; (4) how they differ in citation coverage and traceability ("auditability"); and (5) whether the advantages hold across two hosting set-ups and query difficulty levels. They claim four contributions: a pre-registered evaluation protocol with bootstrap confidence intervals and paired tests, a claim-level auditability metric, instrumented LangChain templates for each paradigm, and decision-support visuals for choosing a pipeline under budget and service-level constraints.

2. The four architectures compared

Pipeline	How it retrieves (paper's configuration)	Strength	Weakness
Sparse	BM25 keyword scoring (k1 = 1.2, b = 0.75); 512-token chunks; top-10, optional cross-encoder re-rank to 5	Every score is inspectable; millisecond lookups on a CPU; precise for standardised wording	Misses paraphrases and other languages; new tickers and SKUs need re-indexing
Dense	all-mpnet-base-v2 embeddings (768-d) in an HNSW index; 384-token chunks; top-20, MiniLM cross-encoder to 5	Finds evidence with no word overlap; works across languages	Similarity scores are opaque to auditors; embedding the corpus is costly and must be redone when it changes
Hybrid	BM25 and dense retrieval run in parallel (15 + 15 passages) and merged with reciprocal-rank fusion, then re-ranked to 8	Precision of BM25 plus recall of embeddings; one track compensates when the other fails	Two indexes to build, tune and keep in sync
Fusion	An LLM writes m = 4 sub-queries, each retrieving 10 passages (or Fusion-in-Decoder over up to 28); 256-token chunks; re-rank to 10	Gathers evidence from several angles; strongest factual coverage	Slowest; larger prompts; harder to trace which passage produced which claim

3. Experimental setup

The finance corpus holds 3,214 SEC 10-K/10-Q filings, investor-relations pages and central-bank circulars from June 2019 to June 2025 (92.7 M tokens), and the e-commerce corpus holds 4,836 product, pricing, returns-policy and help-centre passages in English and Turkish (68.3 M tokens). Both were cleaned, de-duplicated and normalised for currencies and dates, and a separate evidence pool of 7,150 passages carries source IDs and timestamps.

The test set has 500 finance and e-commerce questions, stratified by domain and difficulty. A quarter of them are deliberately unanswerable (built with time shifts and entity perturbations), so a system cannot score well simply by refusing. Annotators agreed substantially on a 20% sample (Cohen's $\kappa = 0.77$ for answerability, 0.69 for domain and 0.74 for difficulty).

Two generators answer the questions: cloud GPT-4o and an on-premises Llama-3-8B-Instruct (4-bit GGUF, llama.cpp on a MacBook Pro M3), both at temperature 0.2 with no fine-tuning. Within each comparison the generator and prompts stay fixed, so any difference can be put down to retrieval. Answers are scored with the RAGAS metrics (faithfulness, the share of answer claims entailed by a cited passage; answer relevance; context precision; and context recall at k), plus auditability (the share of claims with a valid, recent citation) and end-to-end latency, with 95% bootstrap confidence intervals and paired Wilcoxon tests under Holm correction.

4. Key results

Pipeline	Faithfulness	Answer relevance	Context precision	Context recall	Auditability	Median latency
Zero-shot (no retrieval)	0.58	0.61	0.42	0.38	0.35	680 ms
Sparse (BM25)	0.70	0.76	0.66	0.62	0.71	880 ms
Dense (ANN)	0.74	0.80	0.63	0.78	0.73	1,020 ms
Hybrid (RRF + re-rank)	0.80	0.83	0.69	0.75	0.79	1,110 ms
Fusion (expansion / FiD)	0.82	0.84	0.71	0.77	0.81	1,230 ms

Values are means from the paper's Table 3 (95% CIs ±0.02–0.04). Higher is better, except latency.

Without retrieval, GPT-4o invented monetary figures or out-of-date regulatory thresholds in roughly one answer in three, and every RAG pipeline cut that to single digits. Faithfulness rises from 0.58 to between 0.70 and 0.82, and auditability roughly doubles, from 0.35 to 0.71–0.81.

Fusion scores highest on faithfulness, relevance, precision and auditability, but it is also the slowest, at a 1.23 s median that is 550 ms more than no retrieval. Hybrid comes within 0.02 of fusion on nearly every metric while running 120 ms faster, which is why the authors call it the best overall trade-off.

Dense retrieval reaches the highest context recall (0.78) because it matches meaning rather than words. Sparse is the fastest RAG pipeline at 880 ms, but its recall of 0.62 suffers on paraphrased and multilingual questions.

Answers that cite primary sources such as EDGAR filings, or recent passages, are consistently more faithful, so the authority and freshness of the sources moderate how accurate RAG can be.

5. Recommendations, limitations and future work

For deployment, the authors suggest sparse or hybrid retrieval when auditability and strict latency dominate, dense or hybrid for paraphrase-heavy or multilingual interactive traffic, and fusion, ideally with citation filtering, when maximum factual coverage matters and batch or offline latency is acceptable.

The authors acknowledge that only two domains are tested, that results may change with other generators, encoders, indexes and prompt budgets, that RAGAS-style metrics are automatic proxies for human judgement, and that cost and carbon are not measured. For future work they propose dynamic indexing, retrieval-aware decoding and joint training, long-context

generators with claim-level citations, and cross-domain replication with human judges.

6. Critical assessment

The paper is a useful practitioner's guide: the four designs are explained with concrete examples, the configurations are fully specified, and including unanswerable questions is a good safeguard. Its text and radar plot, however, present sparse retrieval as the most auditable and most precise pipeline, while its own Table 3 gives sparse the lowest auditability of the four RAG pipelines (0.71) and a context precision (0.66) behind both hybrid (0.69) and fusion (0.71).

Fusion's faithfulness (0.82 ± 0.02) overlaps hybrid's (0.80 ± 0.02), and the paired-test p-values and effect sizes are deferred to a supplement, so the claim that fusion is the most faithful pipeline is not clearly supported, and hybrid may well be the real winner.

Results are given as one number per pipeline, with no split by generator (GPT-4o against Llama), domain or difficulty, even though RQ5 asks for exactly that, and the sensitivity ablations and Pareto frontiers promised for RQ2 never appear.

Answer keys are not released, scoring is fully automatic, and the code and data are available only on request, so nobody can check the results independently. Editing residue such as "addressing the reviewers' requests" also remains in the published text.

Retrieval clearly and substantially reduces hallucination in fact-critical domains, and hybrid retrieval is the safest default, but finer distinctions between the designs need the stratified results the paper does not yet report.